

Problemformulering

Title: Quantifying relativistic effects of exchanged pions in light nuclei
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Abstract: The goal of the project is to investigate how treating exchanged pions relativistically versus non-relativistically modifies selected observables (binding energy, charge radius, quadrupole moment, and, if feasible, low-energy phase shifts) for a two-nucleon system and, optionally, a three-nucleon system.

Aktivitets- og vejledningsplan

I have meetings with Dimtri planned once a week.

First I have to touch up on quantum mechanics, and read beyond what I have already learned.

I have to find relevant sources, and figure out how to implement the relativistic kinetic energy operator.

Then I will solve what I can analytically, and after that start implementing the code.

In the end i will collect it all in my project.